

The reproducibility paradox of AD blood transcriptomics, and its resolution

The paradox

(observed for >10 years)

Study 1

AUC 0.91

GeneA, GeneB, GeneC

reported markers

Study 2

AUC 0.88

GeneD, GeneE, GeneF

reported markers

Study 3

AUC 0.93

GeneG, GeneH, GeneI

reported markers

High accuracy in every study

...yet gene sets barely overlap, and none replicates clinically.

Why?

The cell-composition account

(this work)

AD

Myeloid-up / lymphoid-down shift
(reproducible across cohorts)

Bulk blood RNA = composition-weighted
average of leukocyte states

ANY sufficiently large set of
myeloid-marker genes classifies AD
(by indirectly counting cells)

Gene lists differ between studies;
underlying signal does NOT.

Different studies select different gene sets — but they are all reading out the same low-dimensional myeloid composition phenotype.